

Synthetic Market Phase 2 Geometry

Dense and minimal scalar sweeps for the counting-manifolds-inspired market-state program. This report compares Phase 2 against the earlier Phase 1 pilot and focuses on whether denser synthetic sweeps make the scalar geometry more manifold-like.

DATE	PROMPTS	FAMILIES	VOLUME
20 March 2026	984 market-only	738 dense, 246 minimal	Dedicated synthetic

MAIN READ

Phase 2 improves the scalar-geometry story, but only partially.

The denser synthetic sweeps materially improved `pct_5m` ordering and noticeably improved concentration geometry in the minimal setting. But the market representation still does not collapse into a clean, universal one-dimensional manifold across all scalar families. The hypothesis survives; it is not yet closed.

PHASE 2 SCALE	BEST REGRESSION R^2	BEST SCALAR GEOM.	PHASE 1 <code>pct_5m</code>
984 market-only prompts	0.997 dense, layer 2	0.741 minimal <code>pct_5m</code> , layer 28	0.637 baseline for comparison

Scope

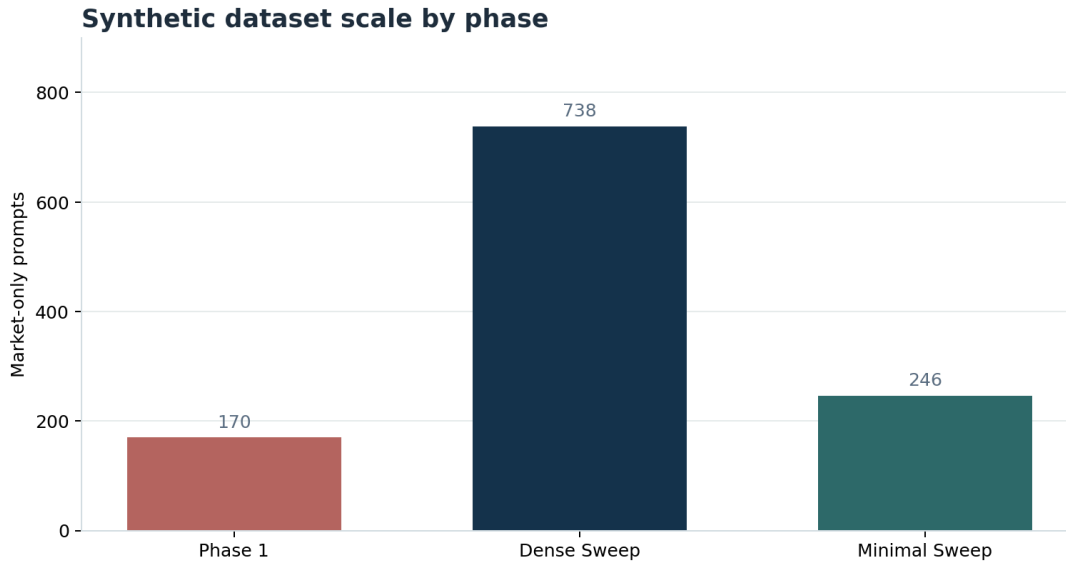
Phase 2 was designed to answer the specific objection left open by Phase 1: perhaps the pilot dataset was simply too coarse to reveal a cleaner scalar geometry. To test that, we expanded the synthetic market-only slice in two directions:

- **Dense sweep.** Repeat scalar values across varied background rosters to test whether the geometry survives more prompt diversity.
- **Minimal sweep.** Strip the prompt down so only the target scalar varies against a flatter background, testing whether a cleaner manifold appears when context clutter is removed.
- **Same pooled states.** Capture and analyze the same `row_mean_i` and `row_eos_i` states as before so the comparison is apples-to-apples.

The result is a stronger geometric read, not just a bigger dataset.

Dataset

Phase 2 contains 984 market-only prompts on the dedicated synthetic Modal volume: 738 dense-sweep prompts and 246 minimal-sweep prompts. This should be read alongside the smaller 170-prompt Phase 1 pilot rather than in isolation.



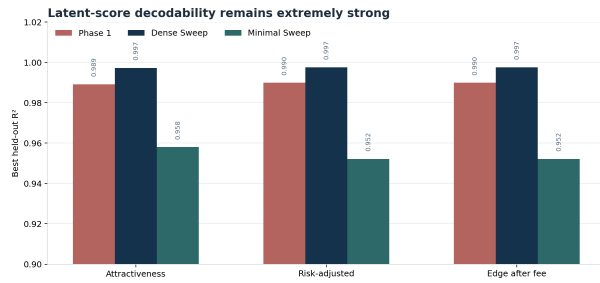
Phase 2 is substantially larger than the Phase 1 pilot, with most of the new scale allocated to denser scalar sweeps rather than more context variants.

Main Quantitative Findings

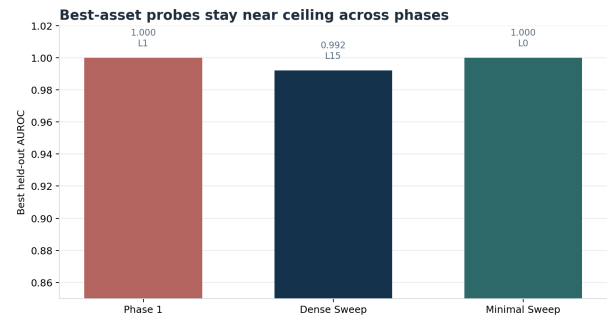
Metric	Phase 1	Dense Sweep	Minimal Sweep
Best attractiveness R^2	0.989	0.997	0.958
Best risk-adjusted R^2	0.990	0.997	0.952
Best edge-after-fee R^2	0.990	0.997	0.952
Best-asset AUROC	1.000	0.992	1.000
Best pct_5m geometry	0.637	0.730	0.741
Best net_flow_5m geometry	0.659	0.565	0.573
Best top20_holder_pct geometry	0.452	0.518	0.602

The broad pattern is clear:

- The latent variables remain very strongly linearly recoverable.
- pct_5m geometry improved substantially in both Phase 2 variants.
- Concentration geometry improved most in the minimal slice.
- net_flow_5m did **not** improve relative to Phase 1, suggesting that not every scalar family wants the same geometric treatment.



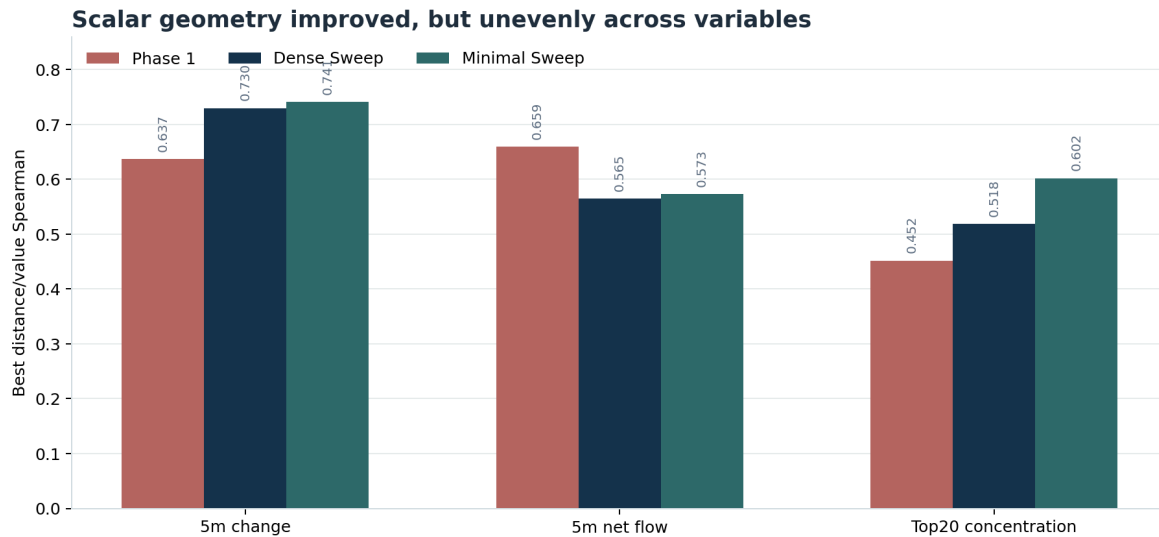
Latent-score regression remains near ceiling. Dense-sweep prompts slightly improve decodability; minimal prompts are a bit harsher but still very strong.



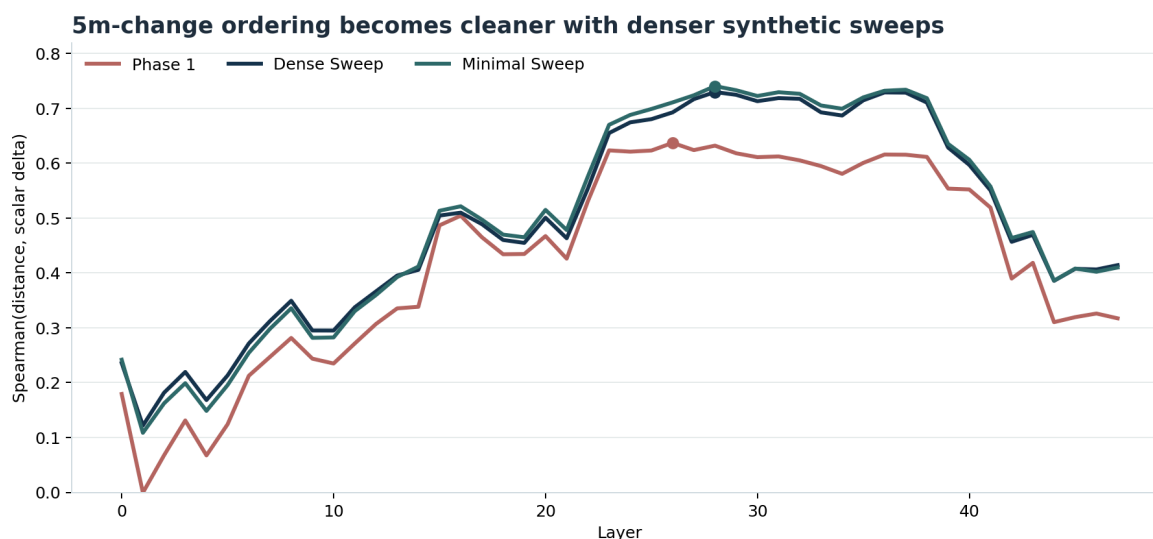
Best-asset probes remain near trivial across all phases, so the real action in Phase 2 is the scalar-geometry comparison, not the choice probes.

Geometry Read

The strongest Phase 2 evidence is not that **everything** got cleaner. It is that some scalar families got meaningfully cleaner under the right synthetic pressure.



The largest Phase 2 win is pct_5m. Concentration also improves, especially in the minimal slice. Net-flow geometry remains more ambiguous and does not improve over Phase 1.



pct_5m becomes substantially more ordered in Phase 2 and peaks in the late 20s. This is the clearest sign that denser scalar sweeps are exposing a more stable geometric organization than the pilot did.

Interpretation

SUPPORTED NOW

Denser synthetic sweeps do improve the scalar-geometry picture. The pilot's weak geometry result was at least partly a data-resolution problem rather than a pure hypothesis failure.

NOT SUPPORTED YET

We still do not have a crisp counting-manifolds-style one-dimensional market scalar manifold that appears uniformly across variable families.

Best current interpretation:

- pct_5m is the cleanest candidate for a counting-manifolds-style synthetic market variable.
- top20_holder_pct may also admit cleaner geometry, but appears more sensitive to prompt simplification than to dense varied backgrounds.
- net_flow_5m is either represented in a less purely scalar way or is more entangled with other latent structure than the current synthetic sweeps capture.
- The model is preserving the right variables very early, but the geometry differs by variable family. That is closer to a **family of manifold-like structures** than to one universal scalar manifold.

Methodological Notes

The dense run emitted many sklearn `LinAlgWarning` messages during ridge regression. These were numerical-conditioning warnings, not capture or analysis failures:

- the job completed successfully and wrote the final JSON
- the warnings arose because the dense synthetic sweeps create highly collinear feature directions
- after this run, the ridge probe path was patched to use the more numerically stable `svd` solver for future runs

That means the current Phase 2 findings are usable, and future reruns should be quieter.

What This Means For The Hypothesis

Phase 2 does **not** fundamentally challenge the hypothesis. It sharpens it.

The cleaner conclusion is:

- there is meaningful scalar organization in the market-row representation
- the clarity of that organization depends on **which scalar** we study and **how much nuisance structure** we leave in the prompt
- the next step should focus less on “one scalar manifold for everything” and more on identifying which market variables admit the cleanest geometric treatment

Next Moves

The highest-value next moves are now more specific:

1. Treat `pct_5m` as the lead variable for the counting-manifolds analogue and deepen the synthetic sweep around it.
2. For `top20_holder_pct`, test whether its cleaner minimal-sweep behavior survives a modest reintroduction of background variation.
3. For `net_flow_5m`, construct more targeted synthetic families rather than assuming it should behave like momentum.
4. Start causal tests on the strongest variable-specific candidates before scaling context ladders again.

This is enough progress to justify a Phase 2 report. It is not yet enough to claim the broader market-manifold story is finished.

Synthetic Market Phase 2 Geometry Report — 20 March 2026. 984 market-only captures on the dedicated synthetic Modal volume. Comparison baseline: Phase 1 pilot with 170 market-only captures.